Baseline Maths Assessment

Year 3

This resource corresponds with the New National Curriculum objectives for Year 2, and has been designed to be used as an aide in assessing pupils' baseline mathematical knowledge at the start of Year 3.

The content addresses each curricular objective outlined in the New National Curriculum in chronological order to give insight into strengths or gaps in each child's mathematical knowledge.

More Assessment resources.

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Baseline Maths Assessment – Year 3

Year 2 National Curriculum Objectives:

Number – number and place value

1. Count in steps of 2, 3, and 5 from 0, and in tens from any number, forward and backward

2. Recognise the place value of each digit in a two-digit number (tens, ones)

3. Identify, represent and estimate numbers using different representations, including the number line

4. Compare and order numbers from 0 up to 100; use <, > and = signs

5. Read and write numbers to at least 100 in numerals and in words

1, 2, 3, 4 & 5. Use place value and number facts to solve problems

Number – addition and subtraction

26 & 27. Solve problems with addition and subtraction using concrete objects and pictorial representations, including those involving numbers, quantities and measures

6. Solve problems with addition and subtraction applying their increasing knowledge of mental and written methods

7. Recall and use addition and subtraction facts to 20 fluently, and derive and use related facts up to 100

8. Add and subtract numbers using concrete objects, pictorial representations, and mentally for a two-digit number and ones

9. Add and subtract numbers using concrete objects, pictorial representations, and mentally for a two-digit number and tens

10. Add and subtract numbers using concrete objects, pictorial representations, and mentally for two two-digit numbers

11. Add and subtract numbers using concrete objects, pictorial representations, and mentally for adding three one-digit numbers

12. Show that addition of two numbers can be done in any order (commutative) and subtraction of one number from another cannot

13. Recognise and use the inverse relationship between addition and subtraction and use this to check calculations and solve missing number problems

Baseline Maths Assessment – Year 3

Year 2 National Curriculum Objectives:

Number – multiplication and division

14. Recall and use multiplication and division facts for the 2, 5 and 10 multiplication tables, including recognising odd and even numbers

15. Calculate mathematical statements for multiplication and division within the multiplication tables and write them using the multiplication (×), division (÷) and equals (=) signs

16. Show that multiplication of two numbers can be done in any order (commutative) and division of one number by another cannot

17. Solve problems involving multiplication and division, using materials, arrays, repeated addition, mental methods, and multiplication and division facts, including problems in contexts

Number – fractions

18, 19, 20 & 21. Recognise, find, name and write fractions 1/3 , 1/4 , 2/4 and 3/4 of a length, shape, set of objects or quantity

22. Write simple fractions for example, 1/2 of 6 = 3 and recognise the equivalence of 2/4 and 1/2

Measurement

23 & 24. Choose and use appropriate standard units to estimate and measure length/height in any direction (m/cm); mass (kg/g); temperature (° C); capacity (litres/ml) to the nearest appropriate unit, using rulers, scales, thermometers and measuring vessels

25. Compare and order lengths, mass, volume/capacity and record the results using >, < and =

26. Recognise and use symbols for pounds (£) and pence (p); combine amounts to make a particular value

26. Find different combinations of coins that equal the same amounts of money

27. Solve simple problems in a practical context involving addition and subtraction of money of the same unit, including giving change

28. Compare and sequence intervals of time

31. Tell and write the time to five minutes, including quarter past/to the hour and draw the hands on a clock face to show these times

29 & 30. Know the number of minutes in an hour and the number of hours in a day

Baseline Maths Assessment – Year 3

Year 2 National Curriculum Objectives:

Geometry – properties of shapes

32. Identify and describe the properties of 2-D shapes, including the number of sides and line symmetry in a vertical line

33. Identify and describe the properties of 3-D shapes, including the number of edges, vertices and faces

34. Identify 2-D shapes on the surface of 3-D shapes, [for example, a circle on a cylinder and a triangle on a pyramid]

(Not covered) Compare and sort common 2-D and 3-D shapes and everyday objects

Geometry – position and direction

35. Order and arrange combinations of mathematical objects in patterns and sequences

36. Use mathematical vocabulary to describe position, direction and movement, including movement in a straight line and distinguishing between rotation as a turn and in terms of right angles for quarter, half and three-quarter turns (clockwise and anticlockwise)

Statistics

37e. Interpret and construct simple pictograms, tally charts, block diagrams and simple tables

37a, b & d. Ask and answer simple questions by counting the number of objects in each category and sorting the categories by quantity

37c. Ask and answer questions about totalling and comparing categorical data

Baseline Maths Assessment – Year 3

Number and Place Value

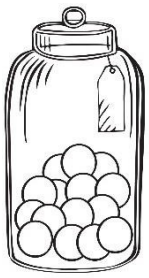
1. Complete the sequence of numbers.

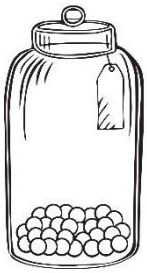
0	2	4	6						
0	3	6	9						
0	5	10	15						
40	50	60	70						
120	110	100	90						

2. Write the value of each underlined digit.

<u>2</u>1	_____	3<u>5</u>	_____	<u>6</u>5	_____
9<u>3</u>	_____	<u>5</u>6	_____	8<u>2</u>	_____

3. Estimate the total number of sweets in each jar pictured below





4. Use the symbols > or < to compare the numbers.

94	☐	76	52	☐	25	54	☐	49

5. Write the following numbers in words.

46	_____	64	_____	17	_____
98	_____	100	_____	30	_____

Addition and Subtraction

6. Solve the problems.

45	+	17	=		87	−	36	=	
63	−	13	=		34	+	43	=	
48	+	23	=		99	−	67	=	

7. Complete the following number bonds mentally.

	+	16	=	20		+	12	=	20
20	−		=	8	20	−		=	2
	+	11	=	20		+	3	=	20
20	−		=	14	20	−		=	7

8. Complete each calculation.

65	+	9	=		87	−	6	=	
43	−	8	=		32	+	9	=	
37	+	7	=		62	−	3	=	

9. Complete each calculation.

49	+	40	=		28	−	10	=	
33	−	20	=		58	+	30	=	
82	+	10	=		76	−	50	=	

10. Complete each calculation.

39	+	40	=	☐	83	−	76	=	☐
59	−	37	=	☐	62	+	26	=	☐
49	+	16	=	☐	78	−	34	=	☐

11. Complete the sums.

6	+	7	+	4	=	☐	3	+	8	+	2	=	☐
9	+	3	+	7	=	☐	9	+	5	+	3	=	☐
5	+	8	+	6	=	☐	4	+	7	+	5	=	☐

12. Write an addition and subtraction sentence for each set of numbers.

34, 17, 17	☐	+	☐	=	☐	☐	−	☐	=	☐
25, 52, 27	☐	+	☐	=	☐	☐	−	☐	=	☐
64, 16, 48	☐	+	☐	=	☐	☐	−	☐	=	☐

13. Solve each missing number problem by using the inverse calculation.

24	+	☐	=	75	−		=	
☐	−	32	=	48	+		=	
☐	+	46	=	97	−		=	

Multiplication and Division

14. Write the next five multiples of each number.
Then, circle the odd numbers in red and the even numbers in blue.

12	14	16	_____	_____	_____	_____	_____
0	5	10	_____	_____	_____	_____	_____
30	40	50	_____	_____	_____	_____	_____

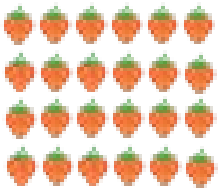
15. Use the symbols x, ÷ or = to complete each number sentence.

5	☐	6	☐	30	70	☐	10	☐	7
12	☐	2	☐	24	16	☐	2	☐	8
45	☐	5	☐	9	40	☐	10	☐	4

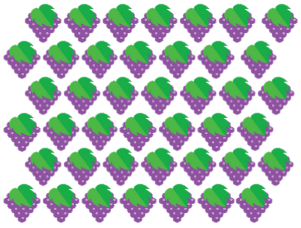
16. Write a multiplication and division sentence for each set of numbers.

10, 50, 5	☐	x	☐	=	☐	☐	÷	☐	=	☐
12, 2, 24	☐	x	☐	=	☐	☐	÷	☐	=	☐
15, 5, 3	☐	x	☐	=	☐	☐	÷	☐	=	☐

17. Write a multiplication sentence, a division sentence, and a repeated addition sentence to describe each array.



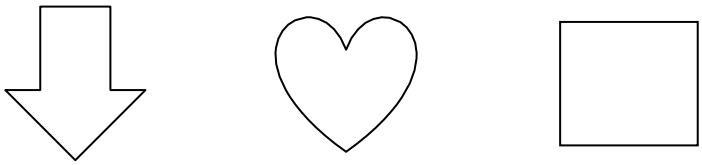
☐ x ☐ = ☐ ☐ ÷ ☐ = ☐



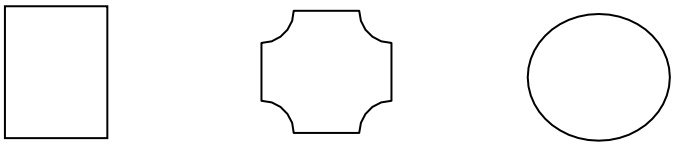
☐ x ☐ = ☐ ☐ ÷ ☐ = ☐

Fractions

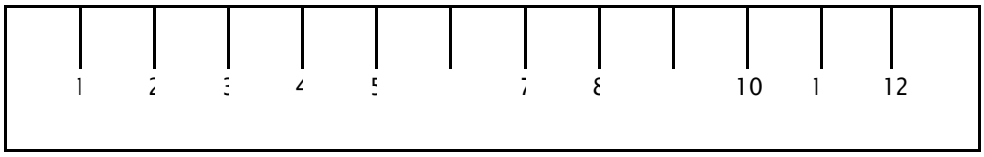
18. Shade $\frac{1}{2}$ of each of these shapes.



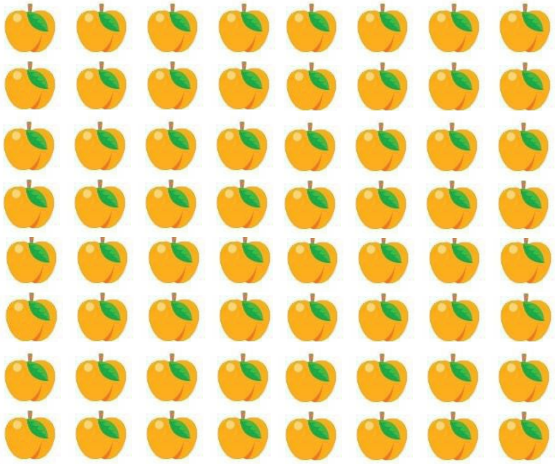
19. Shade $\frac{1}{4}$ of each of these shapes.



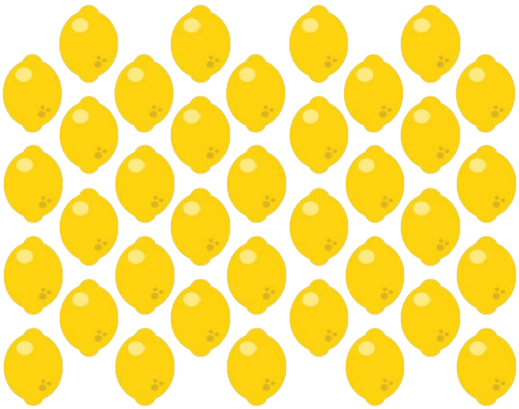
20. Shade $\frac{3}{4}$ of this ruler.



21. Circle $\frac{2}{4}$ of the peaches.



Circle $\frac{1}{4}$ of the lemons.



22. Calculate the given fraction of each whole number.

$\frac{1}{4}$ of 16 = ☐

$\frac{1}{2}$ of 6 = ☐

$\frac{3}{4}$ of 20 = ☐

$\frac{1}{2}$ of 30 = ☐

$\frac{3}{4}$ of 28 = ☐

$\frac{1}{4}$ of 32 = ☐

Measurement

23. Order the following lengths from tallest to shortest.

74cm	25cm	92cm	30cm	101cm

24. Order the following weights from heaviest to lightest.

54kg	29kg	45kg	56kg	39kg

25. Use the <, > or = symbols to compare the following measurements.

34 kg		29 kg	17 l		21 l
25 cm		52 cm	51 m		51 m
26 ˚C		19 ˚C	296 ml		312 ml

26. You have the following coins in your piggy bank.

What is the total amount of money in your piggy bank?

Using only the coins pictured above, show two different ways to make:

26p	45p	£1.10

27. You have 98p. You buy two bags of sweets for 10p each. How much change will you receive? Show your workings in the space below.

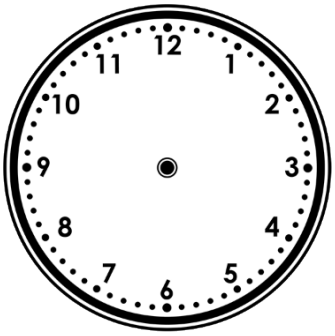
28. Use the <, > or = symbols to compare the following lengths of time.

60 minutes		1 hour	2 hours		60 minutes
Half an hour		45 minutes	30 minutes		1 hour

29. How many minutes are in one hour?

30. How many hours are in one day?

31. Draw hands on the clocks to show these times.



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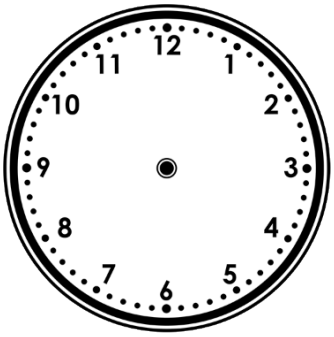

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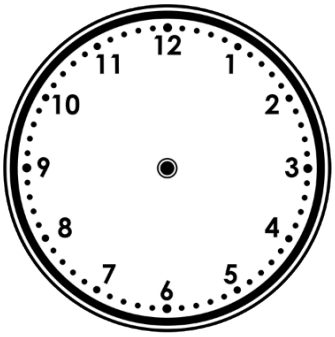

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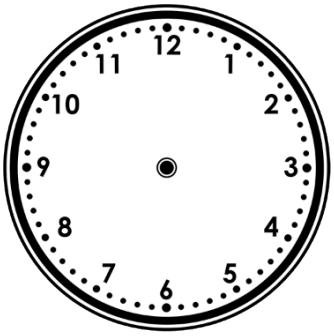

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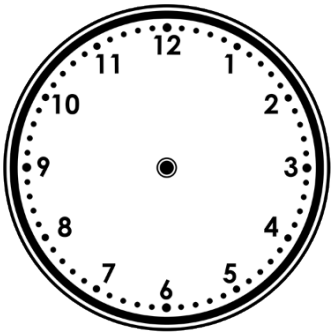

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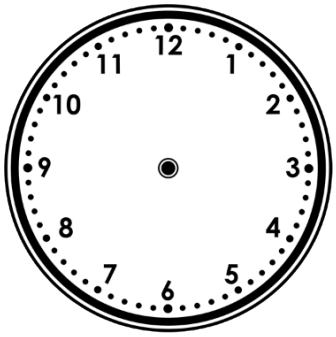
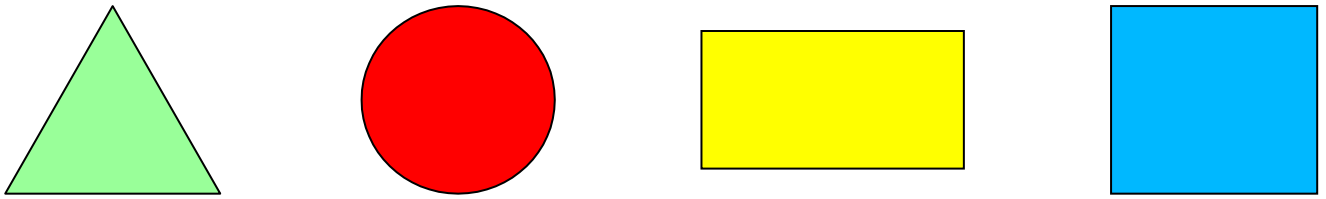


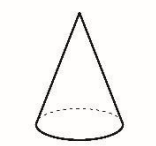
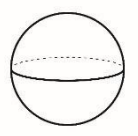
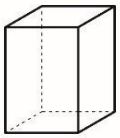
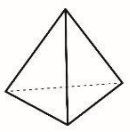
4:45

Geometry

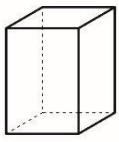
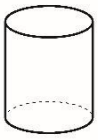
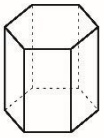
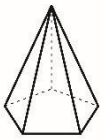
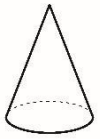
32. Write the number of sides each shape has in the boxes below. Then, draw a vertical line of symmetry through each of the shapes.



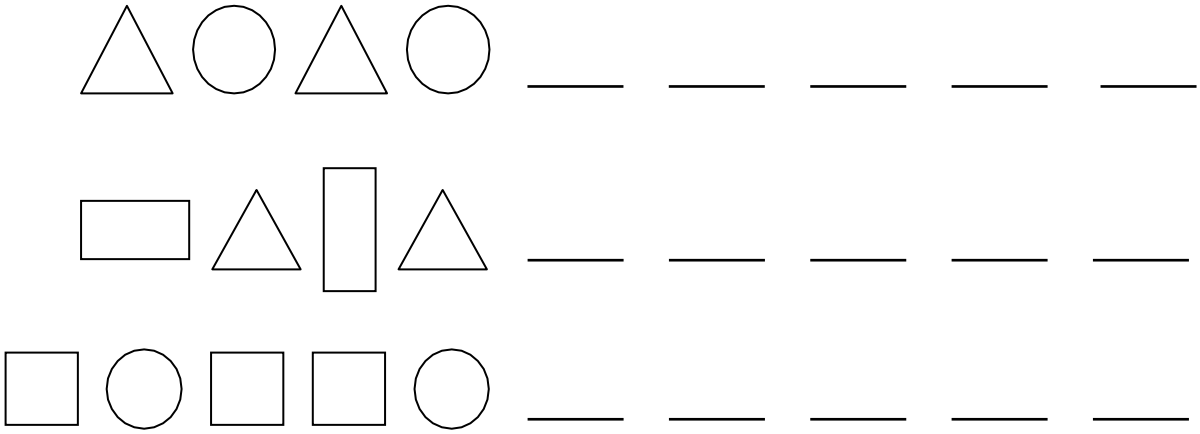
33. Fill in the table with the properties of the 3D shapes.

				
Name of shape				
edges				
vertices				
faces				

34. List any 2D shapes you see on the surface of each of the following 3D shapes.

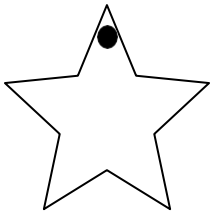
				

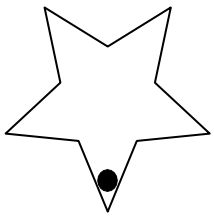
35. Repeat the patterns and complete each sequence.

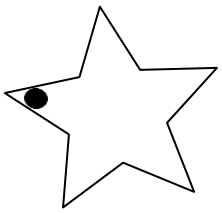


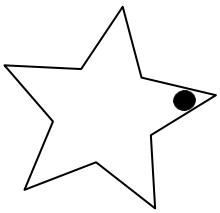
36. Look at the star. Match each shape below to the degree of rotation from the word bank.

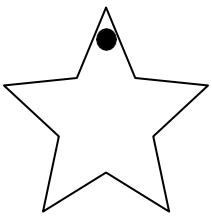
half turn	quarter turn clockwise	whole turn	three-quarter turn clockwise





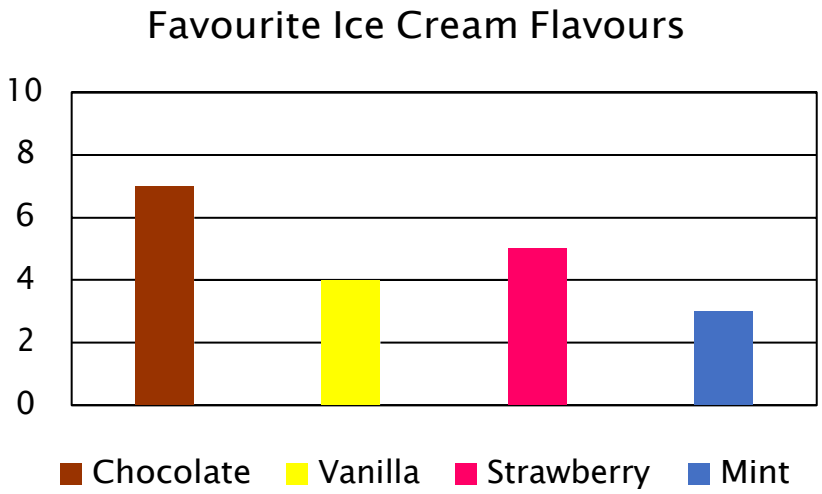






Statistics

37. Read the bar chart and answer the questions below.



a. What is the most popular ice cream flavour?

b. What is the least popular ice cream flavour?

c. How many people voted in total?

d. List the four flavours from most to least popular.

e. Fill in the tally chart below to shows the number of votes for each flavour.

Baseline Maths Assessment - Year 3

Number and Place Value

1. 8, 10, 12, 14, 16, 18
   12, 15, 18, 21, 24, 27
   20, 25, 30, 35, 40, 45
   80, 90, 100, 110, 120, 130
   80, 70, 60, 50, 40, 30

2. 2 tens 5 ones 6 tens
   3 ones 5 tens 2 ones

3. 4 rows of 4 sweets = 16, 4 rows of 8 sweets = 32, 5 rows of 10 sweets = 50

4. > , > , >

5. forty-six sixty-four seventeen
   ninety-eight one hundred thirty

Addition and Subtraction

6. 62 51
   50 77
   71 32

7. 4 8
   12 18
   9 17
   6 13

8. 74 81
   35 41
   44 59

9. 89 18
   13 88
   92 26

10. 79 7
    22 88
    65 44

11. 17 13
    19 17
    19 16

12. 17 + 17 = 34, 34 - 17 = 17
25 + 27 = 52 or 27 + 25 = 52, 52 - 25 = 27 or 52 - 27 = 25
48 + 16 = 64 or 16 + 48 = 64, 64 - 16 = 48 or 64 - 48 = 16

13. 51 75 - 24 = 51
80 48 + 32 = 80
51 97 - 46 = 51

Multiplication and Division

14.

18	20	22	24	26
15	20	25	30	35
60	70	80	90	100

15.

5	x	6	=	30		70	÷	10	=	7
12	x	2	=	24		16	÷	2	=	8
45	÷	5	=	9		40	÷	10	=	4

16. 10 x 5 = 50 or 5 x 10 = 50, 50 ÷ 10 = 5 or 50 ÷ 5 = 10
12 x 2 = 24 or 2 x 12 = 24, 24 ÷ 2 = 12 or 24 ÷ 12 = 2
3 x 5 = 15 or 5 x 3 = 15, 15 ÷ 5 = 3 or 15 ÷ 3 = 5

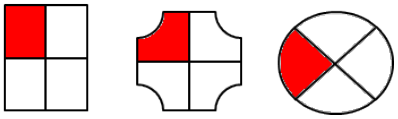
17. 4 x 6 = 24 24 ÷ 4 = 6
6 + 6 + 6 + 6 = 24 or 4 + 4 + 4 + 4 + 4 + 4 = 24

6 x 7 = 42 42 ÷ 7 = 6
7 + 7 + 7 + 7 + 7 + 7 = 42 or 6 + 6 + 6 + 6 + 6 + 6 + 6 = 42

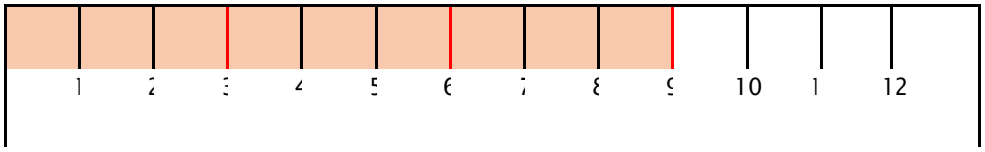
Fractions

18.

19.



20.



21. Circle any 32 Circle any 9

22. 4 15
3 21
15 8

Measurement

23. 101cm | 92cm | 74cm | 30cm | 25cm

24. 56kg | 54kg | 45kg | 39kg | 29kg

25. > <
< =
> <

26. £4.78
26p: 10p + 10p + 5p + 1p / 20p + 5p + 1p / 10p + 5p + 5p + 5p + 1p
45p: 20p + 20p + 5p / 20p + 10p + 10p + 5p / 20p + 10p + 5p + 5p + 5p
£1.10: £1 + 10p / £1 + 5p + 5p / 50p + 50p + 10p / 50p + 50p + 5p + 5p
50p + 20p + 20p + 10p + 10p / 50p + 20p + 20p + 10p + 5p + 5p

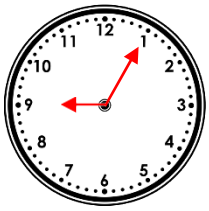
27. 10p + 10p = 20p, 98p – 20p = 78p

28. = >
< <

29. 60

30. 24

31.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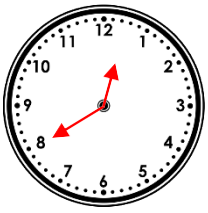
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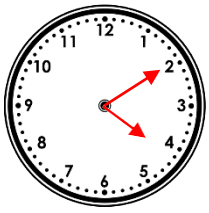
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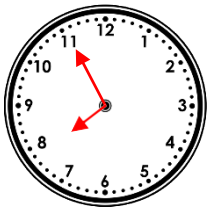
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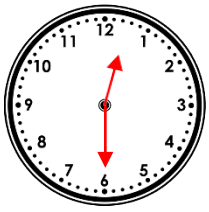
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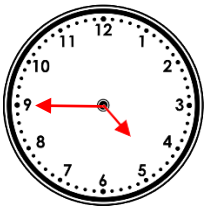

4:45

Geometry

32. 3 sides, 1 side, 4 sides, 4 sides, ensure lines of symmetry are in the centre.

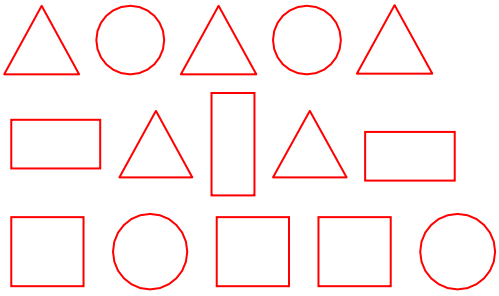
33.

Name of shape	cone	sphere	cuboid	pyramid (tetrahedron)
edges	1	0	12	6
vertices	1	0	8	4
faces	2	1	6	4

34.

rectangle	circle	rectangle	triangle	circle
square		hexagon	pentagon	

35.



36. half turn / three-quarter turn clockwise / quarter turn clockwise / whole turn

37 a. chocolate
b. mint
c. 19
d. chocolate, strawberry, vanilla, mint
e.

Flavour	Number of votes
Chocolate	~~||||~~ ||
Vanilla	~~||||~~
Strawberry	||||
Mint	|||